

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA1610 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

QUESTIONS: 05

Total Marks:50

Annexure A

MOCK INTERVIEW

<i>Criteria</i>	Technical Knowledge (2.5)	Problem Solving (2.5)	Analytical Thinking (1.5)	<i>Communication(1.5)</i>	Confidence (1)	Response to Questions(1)	<i>Total(10)</i>
<i>Weightage</i>	25%	25%	15%	15%	10%	10%	100%
<i>Q1</i>							
<i>Q2</i>							

Code of Conduct:

I Heisenberg Raja G B (192324158) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Heisenberg Raja

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

Annexure A

Mock Interview – Sample Questions (5 Questions)

Q1. Dataset: Retail Transactions

TID	Items
T1	Milk, Bread
T2	Milk, Butter
T3	Bread, Butter
T4	Milk, Bread, Butter
T5	Milk, Eggs

Question:

Based on the dataset, explain how association rule mining can improve sales in a retail store. Provide a practical example.

Q2. Dataset: Transaction Data

TID	Items
T1	A, B
T2	B, C
T3	A, C
T4	A, B, C
T5	B, C

Question:

Using this dataset, explain the differences between Apriori and FP-Growth algorithms and when each should be used.

Q3. Dataset: Transaction Data

TID	Items
T1	A, B
T2	A
T3	B
T4	A, B
T5	A

Question:

Explain support and confidence using the dataset and describe their importance in association rule mining.

Q4 . Dataset: Multilevel Data

TID	Items
T1	Electronics, Mobile
T2	Electronics, Laptop
T3	Electronics, Mobile
T4	Electronics, Mobile
T5	Electronics, Laptop

Question:

Explain multilevel association rules using the dataset and compare them with multidimensional

Q5. Dataset: Large Transaction Sample

TID	Items
T1	A, B, C
T2	A, B
T3	B, C
T4	A, C
T5	A, B, C

Question:

Discuss challenges in mining large datasets using the given sample and explain how scalable methods address these issues.

Q1. Dataset: Retail Transactions

TID	Items
T1	Milk, Bread
T2	Milk, Butter
T3	Bread, Butter
T4	Milk, Bread, Butter
T5	Milk, Eggs

Question:

Based on the dataset, explain how association rule mining can improve sales in a retail store. Provide a practical example.

Association Rule Mining in Retail

Association rule mining finds relationships between products that are frequently purchased together. Retail stores can use these relationships for cross-selling, product placement, recommendations, and promotions.

From the dataset:

- Milk appears in T1, T2, T4, T5 → $4/5 = 80\%$ **support**
- Bread appears in T1, T3, T4 → $3/5 = 60\%$ **support**
- Butter appears in T2, T3, T4 → $3/5 = 60\%$ **support**
- Milk and Bread occur together in T1 and T4 → $2/5 = 40\%$ **support**

For the rule **Milk** → **Bread**:

$$\text{Confidence} = \text{Support (Milk, Bread)} / \text{Support(Milk)}$$
$$= (2/5) / (4/5) = 50\%$$

This means that among customers who buy Milk, **50% also buy Bread**.

Practical example:

The store could place bread close to milk or offer a Milk + Bread combo discount. It could also recommend bread to a customer buying milk online. This can increase cross-selling and overall sales.

Interview point: Association rules help retailers understand customer buying behaviour rather than relying only on intuition.

Q2. Dataset: Transaction Data

TID Items

T1 A, B
T2 B, C
T3 A, C
T4 A, B, C
T5 B, C

Question:

Using this dataset, explain the differences between Apriori and FP-Growth algorithms and when each should be used.

Apriori vs FP-Growth

Both **Apriori** and **FP-Growth** are used to find frequent item sets, but they work differently.

Apriori

Apriori generates candidate item sets and repeatedly scans the database.

For example, from the dataset:

- A occurs in T1, T3, T4 → $3/5 = 60\%$
- B occurs in T1, T2, T4, T5 → $4/5 = 80\%$
- C occurs in T2, T3, T4, T5 → $4/5 = 80\%$
- B,C occurs in T2, T4, T5 → $3/5 = 60\%$

Apriori uses the principle that **if an itemset is infrequent, all of its supersets must also be infrequent**. This helps prune unnecessary candidates.

FP-Growth

FP-Growth does not generate a large number of candidate item sets. Instead, it builds an **FP-tree**, which compresses the transaction database, and then mines frequent patterns from the tree.

Comparison

Feature	Apriori	FP-Growth
Candidate generation	Yes	No
Database scans	Multiple	Usually fewer
Memory	Can require more	FP-tree requires memory
Speed on large data	Generally slower	Generally faster
Implementation	Simpler to understand	More complex

When to use:

- Use Apriori when the dataset is relatively small and simplicity is important.
- Use FP-Growth for large datasets with many transactions, where candidate generation would become expensive.

Interview Point:

“Apriori is easier to understand but can be computationally expensive because of candidate generation and repeated database scans. FP-Growth is generally more efficient because it compresses the database into an FP-tree and avoids candidate generation.

Q3. Dataset: Transaction Data

TID Items

T1 A, B

T2 A

T3 B

T4 A, B

T5 A

Question:

Explain support and confidence using the dataset and describe their importance in association rule mining.

Support and Confidence

The dataset contains 5 transactions:

- A occurs in T1, T2, T4, T5 → **4 transaction**
- B occurs in T1, T3, T4 → **3 transactions**
- A and B occur together in T1 and T4 → **2 transactions**

Support

Support tells us **how frequently an itemset occurs in the entire dataset.**

For {A, B}:

$$\text{Support(A,B)} = 2/5 = 40\%$$

So, 40% of all transactions contain both A and B.

Confidence

Confidence tells us how often B is purchased when A is purchased.

For the rule **A → B**:

$$\text{Confidence(A} \rightarrow \text{B)} = \text{Support(A,B)} / \text{Support(A)}$$

$$= (2/5) / (4/5)$$

$$= 50\%$$

Therefore, **50% of customers who bought A also bought B.**

For comparison, for **B → A**:

$$\text{Support(A,B)} = 2/5$$

$$\text{Support(B)} = 3/5$$

$$\text{Confidence} = (2/5) / (3/5) = 66.67\%$$

Importance

- **Support** helps identify patterns that occur frequently enough to be useful.
- **Confidence** measures the reliability of an association rule.
- Both help eliminate rules that are either too rare or unreliable.

Interview point:

A high confidence rule is not automatically useful; in practice, we may also examine **lift** to determine whether the relationship is genuinely stronger than expected by chance.

Q4 . Dataset: Multilevel Data

TID	Items
T1	Electronics, Mobile
T2	Electronics, Laptop
T3	Electronics, Mobile
T4	Electronics, Mobile
T5	Electronics, Laptop

Question:

Explain multilevel association rules using the dataset and compare them with multidimensional

Multilevel Association Rules vs Multidimensional Association Rules

Multilevel association rules discover relationships at different levels of abstraction in a concept hierarchy.

In this dataset:

- Level 1: **Electronics**
- Level 2: **Mobile / Laptop**

For example:

Electronics → Mobile

Mobile occurs in T1, T3 and T4:

Support = $3/5 = 60\%$

Laptop occurs in T2 and T5:

Support = $2/5 = 40\%$

A multilevel analysis could therefore identify patterns such as:

- Customers frequently buy Electronics
- Within Electronics, Mobile is more frequently purchased than Laptop

The key idea is that the same data can be analysed at different levels of detail.

Multilevel vs Multidimensional

Feature	Multilevel	Multidimensional
Focus	Different abstraction levels	Different attributes/dimensions
Example	Electronics → Mobile	Age → Income → Product
Hierarchy	Usually uses concept hierarchy	Uses multiple attributes
Purpose	Analyse patterns at different granularities	Analyse relationships across attributes

Interview Point:

“Multilevel association rules look at the same type of information at different levels, such as Electronics → Mobile. Multidimensional association rules combine different dimensions or attributes, such as Customer Age, Location, Income, and Product.

Q5. Dataset: Large Transaction Sample

TID Items

- T1 A, B, C
- T2 A, B
- T3 B, C
- T4 A, C
- T5 A, B, C

Question:

Discuss challenges in mining large datasets using the given sample and explain how scalable methods address these issues.

Challenges in Mining Large Datasets

Even though the sample contains only 5 transactions, it demonstrates the types of patterns that become challenging when the dataset becomes very large.

For example, the dataset contains combinations such as:

- A,B,C
- A,B

- B,C
- A,C

As the number of items increases, the possible number of item combinations grows rapidly. This creates several challenges.

Major challenges

1. **Large search space**
With many items, the number of possible itemsets can become extremely large.
2. **High computational cost**
Searching through millions or billions of transactions requires significant processing power.
3. **Repeated database scans**
Algorithms such as basic Apriori may need multiple scans of the database.
4. **Memory limitations**
Large transaction datasets may not fit entirely into memory.
5. **I/O and processing time**
Reading and processing huge datasets can become a major bottleneck.

How scalable methods solve these problem

FP-Growth reduces candidate generation by constructing an FP-tree.

Apriori pruning removes item sets that cannot possibly become frequent.

Partitioning divides a large dataset into smaller portions that can be processed separately.

Sampling can analyse a representative subset when an approximate answer is acceptable.

Parallel and distributed processing allows frameworks such as distributed data-processing systems to divide the workload across multiple machines.

Interview answer:

“The main challenge in large-scale association mining is the explosion of possible item combinations and the computational cost of processing huge transaction databases. Scalable algorithms such as FP-Growth, together with pruning, partitioning, sampling, and distributed processing, reduce computation and allow frequent-pattern mining to work efficiently on large datasets.”

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